

Assignment 10

/* Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle,

Perimeter = $2 * (L + W)$ and Area = $L * W$. */

--drop procedure if exists rectangle;

delimiter \$\$

create procedure rectangle(length decimal(9,2), width decimal(9,2))

begin

declare perimeter decimal(9,2);

declare area decimal(9,2);

set perimeter = $2 * (length + width)$;

set area = length * width;

select perimeter as 'Perimeter', area as 'Area';

end;

\$\$

delimiter ;

```
D6_99024_tejas>call rectangle(3,4);
+-----+-----+
| Perimeter | Area |
+-----+-----+
|      14.00 | 12.00 |
+-----+-----+
1 row in set (0.00 sec)
```

```
/* 2. Write a procedure that declares an integer variable called num, assigns
a value to it, computes and inserts into the temp table the value of the
variable itself, its square, and its cube.*/

drop procedure if exists num_square_cube;

delimiter $$

create procedure num_square_cube(num int)

begin

declare snum int;

declare cnum int;

set snum =num*num;

set cnum=num*num*num;

insert into temp(num, snum, cnum) values(num, snum, cnum);

end;

$$

delimiter ;

-- call num_square_cube(5);
```

```
D6_99024_tejas>select * from temp;
+-----+-----+-----+
| num  | snum | cnum |
+-----+-----+-----+
|  10  |  100 | 1000 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```
drop procedure if exists temp_conversion;

delimiter $$

create procedure temp_conversion(temp float, unit char(1))

begin

declare result float;

if unit = 'f'

then set result = (temp - 32) * 5/9;

elseif unit = 'c'

then set result = (9/5 * temp) + 32;

end if;

select result as 'Converted Temperature';

end;

$$

delimiter ;
```

```
D6_99024_tejas>call temp_conversion(25,'f');
+-----+
| Converted Temperature |
+-----+
|          -3.88889 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

D6_99024_tejas>call temp_conversion(25,'c');
+-----+
| Converted Temperature |
+-----+
|              77 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
drop procedure if exists inch_conversion;
```

```
delimiter $$
```

```
create procedure inch_conversion(inches int)
```

```
begin
```

```
    declare yards int;
```

```
    declare feet int;
```

```
    declare inches_remaining int;
```

```
    set yards = inches / 36;
```

```
    set feet = (inches % 36) / 12;
```

```
    set inches_remaining = inches % 12;
```

```
    select yards as 'Yards', feet as 'Feet', inches_remaining as 'Inches';
```

```
end;
```

```
$$
```

```
delimiter ;
```

```
D6_99024_tejas>call inch_conversion(100);
+-----+-----+-----+
| Yards | Feet | Inches |
+-----+-----+-----+
|      3 |    2 |      4 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
drop procedure if exists check_product;

delimiter $$

create procedure check_product(num1 float, num2 float)
begin
declare product float;
set product = num1 * num2;
if product >= 100 then
    select 'The product is equal to or greater than 100' as 'Result';
else
    select 'The product is less than 100' as 'Result';
end if;
end;

$$

delimiter ;
```

```
D6_99024_tejas>call check_product(14,7);
+-----+
| Result |
+-----+
| The product is less than 100 |
+-----+
1 row in set (0.00 sec)
```